

Hyemin Gu

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Amherst, MA 01002, USA

RESEARCH INTEREST

Mathematically Grounded Machine Learning; Generative Modeling; Stable Learning Algorithms; Dynamical Systems; Scientific Applications of Machine Learning

EMPLOYMENT

- **University of Massachusetts Amherst** *Oct 2025 – Present*
Amherst, MA
Postdoctoral Researcher
 - Advisor: Markos A. Katsoulakis
 - Developing mathematically grounded machine learning architectures for complex learning and optimization, and dynamical system simulators for supply-chain network systems under DARPA's The Right Space Program.
 - Leading the mathematical formulation and algorithmic development of gradient flow-based generative models for continuous prediction of single-cell dynamics.
- **Ewha Womans University Seoul Hospital** *Jul 2020 – Dec 2020*
Seoul, South Korea
Statistics Specialist
 - Developed an R-based Bioconductor pipeline (e.g., limma, edgeR, TCGAblinks) for gene expression analysis, including survival analysis and pathway enrichment
 - Authored a tutorial book on reproducible gene expression workflows for biomedical researchers [[R/Bioconductor tutorial](#)]
 - Conducted statistical training sessions for clinicians, enhancing research reproducibility and data literacy

EDUCATION

- **University of Massachusetts Amherst** *Sep 2020 - Feb 2026*
Amherst, MA
Ph.D. in Applied Mathematics
 - Advisor: Markos A. Katsoulakis
 - Dissertation: [Robust Generative Modeling with Wasserstein Proximal Regularization](#)
- **Ewha Womans University** *Mar 2018 - Feb 2020*
Seoul, South Korea
M.S. in Applied Mathematics
 - Thesis: Convolutional Neural Network for 2D Flow Estimation Problem
- **Ewha Womans University** *Mar 2014 - Feb 2018*
Seoul, South Korea
B.S. in Mathematics and Computational science, Minor in Statistics
 - Thesis: Low cost training of a classification Neural Network with respect to Weight Selection

- [P.4] T Gamage, H Gu, Z Zhang, Z Chen, MA. Katsoulakis, L Rey-Bellet. (2026). **Fine-Tuning Generative Models for Extreme Events via CVaR-Penalized Wasserstein Gradient Flows**.
- [P.3] H Gu, M Tyrrell, T Sahai, MA. Katsoulakis. (2026). **Sharp Stability Threshold and Certification for Designing Stable Residual Architectures**.
- [J.3] YC Cheng, H Gu, et al. (2026). **PROFET Predicts Continuous Gene Expression Dynamics from scRNA-seq Data to Elucidate Heterogeneity of Cancer Treatment Responses**. *Cell Systems*. DOI: <https://doi.org/10.1016/j.cels.2026.101710>
- [P.2] Z Zhang, H Gu, MA. Katsoulakis, T Sahai, et al. (2026). **ISOMORPH: A Supply Chain Digital Twin for Simulation, Dataset Generation, and Forecasting Benchmarks**.
- [J.2] Z Chen, H Gu, MA Katsoulakis, L Rey-Bellet, W Zhu. (2025). **Robust Generative Learning with Lipschitz-Regularized α -Divergences Allows Minimal Assumptions on Target Distributions**. *Information and Inference: A Journal of the IMA*, Volume 14, Issue 4, December 2025, iaaf028. DOI: <https://doi.org/10.1093/imaiai/iaaf028>
- [P.1] H Gu, MA Katsoulakis, L Rey-Bellet, BJ Zhang. (2024). **Combining Wasserstein-1 and Wasserstein-2 proximals: robust manifold learning via well-posed generative flows**.
- [J.1] H Gu, P Birmpa, Y Pantazis, L Rey-Bellet, MA Katsoulakis. (2024). **Lipschitz-Regularized Gradient Flows and Generative Particle Algorithms for High-Dimensional Scarce Data**. *SIAM Journal of Mathematics of Data Science*, 6 (4), 1205-1235. DOI: <https://doi.org/10.1137/23M1587841>

RESEARCH EXPERIENCE

• University of Massachusetts Amherst

Oct 2025 – Present

Postdoctoral Research

Amherst, MA

◦ Primary Research: **DARPA The Right Space (TRS) Program**

TRAGOS: Developing (i) machine learning architectures for complex learning and optimization problems through principled mathematical formulations and (ii) dynamical system simulators for supply-chain network systems.

- * Developed a principled methodology for constructing stable residual neural architectures by establishing necessary and sufficient conditions through optimal-control and Hamilton–Jacobi formulations.
- * Developed ISOMORPH, a Markov-chain-based digital twin of a multi-echelon logistics network for scalable simulation and benchmark dataset generation, and built an [interactive demonstration platform](#) for forecasting research.

◦ Major Collaboration: **Gradient Flow-Based Generative Modeling for Single-Cell Dynamics**

- * Led the mathematical and algorithmic development of PROFET, a gradient flow-based generative framework for continuous trajectory inference from sparse scRNA-seq data, achieving a 2.6–12.5× reduction in prediction error compared with 10 state-of-the-art methods.

◦ Research Collaboration: **Sequential Refinement for Learning Heavy-Tailed Distributions**

- * Contributed to the algorithmic development of a CVaR-penalized Wasserstein gradient flow framework for fine-tuning generative models toward rare-event and heavy-tailed regimes; proposed the initial algorithmic implementation, identified the CVaR-difference refinement strategy, and refined the algorithm to improve robustness.

- Primary Research: **Robust Generative Modeling with Wasserstein Proximal Regularization**
 - * Developed gradient flow-based generative algorithms based on sequential particle transport using Wasserstein-1-regularized f -divergences, and extended the framework to latent-space dynamics for scalable high-dimensional learning.
 - * Developed well-posed flow-based generative models using Wasserstein-1 and Wasserstein-2 proximal-regularized f -divergences for robust learning of distributions supported on low-dimensional manifolds.
- Research Validation: **Cross-Architecture Benchmarking of Wasserstein-Proximal Regularization**
 - * Implemented and benchmarked Wasserstein-1 and Wasserstein-2 proximal regularization across diffusion models, normalizing flows, and GANs, demonstrating the data-agnostic robustness of Wasserstein-1 proximal regularization.
- Research Collaboration: **Gradient Flow-Based Generative Modeling for Single-Cell Dynamics**
 - * Developed the mathematical formulation and core algorithmic framework of PROFET, a gradient flow-based generative model for continuous trajectory inference from sparse scRNA-seq data.

INVITED TALKS

- **Robust Mamba-Based TSF via Decomposed Lipschitz Regularization** at *SIAM Conference on Uncertainty Quantification (UQ)* in Minneapolis, MN, Mar 2026
- **Generative Particle Flows with Force-Matching for Reconstructing Nonlinear Cellular Dynamics** at *SIAM NNP Sectional Annual Meeting* in University Park, PA, Nov 2025
- **Combining Wasserstein-1 and Wasserstein-2 proximals: robust manifold learning via well-posed generative flows** at *Joint Mathematics Meetings – MAA* in Seattle, WA, Jan 2025
- **Wasserstein Proximals Stabilize Training of Generative Models and Learn Manifolds** at *Mathematics of Data Science – SIAM* in Atlanta, GA, Oct 2024

POSTER PRESENTATIONS

- **Data-Driven Force-Matching and Generative Particle Flows for Reconstructing Nonlinear Dynamics from Sparse scRNA-Seq Time Series** at *Sampling, Inference, and Data-Driven Physical Modeling in Scientific Machine Learning – IPAM* in Los Angeles, CA, Jul 2025
- **Lipschitz Regularized Gradient Flows and Latent Generative Particles** at *Mathematics of Data Science – SIAM* in Atlanta, GA, Oct 2024
- **Combining Wasserstein-1 and Wasserstein-2 proximals: robust manifold learning via well-posed generative flows** at *Robust Optimization and Simulation of Complex Stochastic Systems – ICERM* in Providence, RI, Sep 2024
- **Lipschitz Regularized Gradient Flows and Latent Generative Particles** at *Optimal Transport in Data Science – ICERM* in Providence, RI, May 2023
- **Training a 2 layer Neural Network using SVD-generated weights** at *Joint Mathematics Meetings – MAA* in San Diego, CA, Jan 2018
- **Necessary and sufficient conditions for shortest vectors in lattices of dimension 2 and 3** at *Joint Mathematics Meetings – MAA* in Atlanta, GA, Jan 2017

HONORS AND AWARDS

- **Distinguished Thesis Award** Apr 2026
Department of Mathematics and Statistics, University of Massachusetts
- **Graduate Student Travel Grants** Jan 2025
American Mathematical Society
- **Anne and Peter Costa Graduate Prize in Applied Mathematics** Apr 2024
Department of Mathematics and Statistics, University of Massachusetts

PROFESSIONAL SERVICE

- **Journal Reviewer**. *SIAM Journal on Scientific Computing (SISC)*, 2026

TEACHING EXPERIENCE

- **University of Massachusetts Amherst** Feb 2021 – Dec 2024
Graduate Teaching Assistant Amherst, MA
 - (MATH456) **Mathematical Modeling – Mathematics of Generative AIs**: Graded assignments for an advanced modeling course focusing on generative AI methods. (Fall 2024)
 - (MATH532H) **Nonlinear Dynamics and Chaos**: Graded assignments and led [Python tutorials for ODE solving](#). (Fall 2021)
 - (MATH545) **Linear Algebra for Applied Mathematics**: Graded assignments, held office hours, and conducted discussion sessions. (Summer 2021)
 - (MATH235) **Linear Algebra**: Graded assignments. (Spring 2021)
- **Ewha Womans University** Mar 2018 – Dec 2019
Graduate Teaching Assistant Seoul, South Korea
 - **Numerical Differential Equations**: Covered ODE/PDE numerics, Monte Carlo methods, and optimization. (Fall 2019)
 - **Finite Mathematics and Programming**: MATLAB programming, logic, and combinatorics. (Spring 2019)
 - **Calculus 2**: Multivariate calculus. (Fall 2018)
 - **Numerical Analysis**: Solving linear systems, power method, numerical integration/differentiation. (Spring 2018)
 - Graded assignments and held office hours across all courses.

LEADERSHIP EXPERIENCE

- **Department seminar organizer as a part of Graduate Student Advisory Committee** Sep 2022 - May 2025
Department of Mathematics and Statistics, University of Massachusetts - Amherst
 - **Invited speakers** among faculties in the department for introducing their research interests to early-career graduate students and **hosted talks**.
 - **Participated in Graduate Student Advisory Committees meetings**, reported the progress, and discussed future directions.
- **Graduate Student Panelist in Open House for Accepted Students** April 2025
Department of Mathematics and Statistics, University of Massachusetts - Amherst
 - **Represented the graduate student body** by sharing perspectives on academic life, research, and department culture with newly admitted PhD and MS students.

- **Panelist in SIAM Graduate School Panel**

Oct 2024

Department of Mathematics and Statistics, University of Massachusetts - Amherst

- **Led a Q&A panel session** to introduce graduate studies in Mathematics to undergraduate mathematics majors.

- **Graduate student representative in Math Majors Fair**

April 2024

Department of Mathematics and Statistics, University of Massachusetts - Amherst

- **Advised undergraduate mathematics majors** on pursuing graduate studies by sharing insights into graduate life and personal research experiences.

WORKSHOPS AND TRAINING PROGRAMS

- **Industrial Mathematics Academy**

Jun 2018

National Institute for Mathematical Sciences, South Korea

- Presented final results of a group project addressing an industrial problem.
- Proposed a **Convolutional Neural Network** to classify infected individuals from medical images.
- Coordinated team efforts and participated in tutorials on **Python data analysis** and **Keras**, and lectures on **matrix-based data analysis** and **linear programming**.

- **Industrial Mathematics Academy**

Dec 2017

National Institute for Mathematical Sciences, South Korea

- Proposed a **Poisson process-based model** for assessing driving safety scores from On-board Diagnostic data.
- Attended tutorials on the **fundamentals of neural networks**.